H2 Chemistry 9647 Preliminary Examinations 2013 Suggested Answers

Paper 1: Multiple Choice Questions

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
B	D	A	D	C	C	D	D	C	A
Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
C	B	A	D	D	A	B	D	C	B
Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30
B	A	B	C	D	C	D	A	C	D
Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40
D	B	B	A	D	C	B	B	A	A

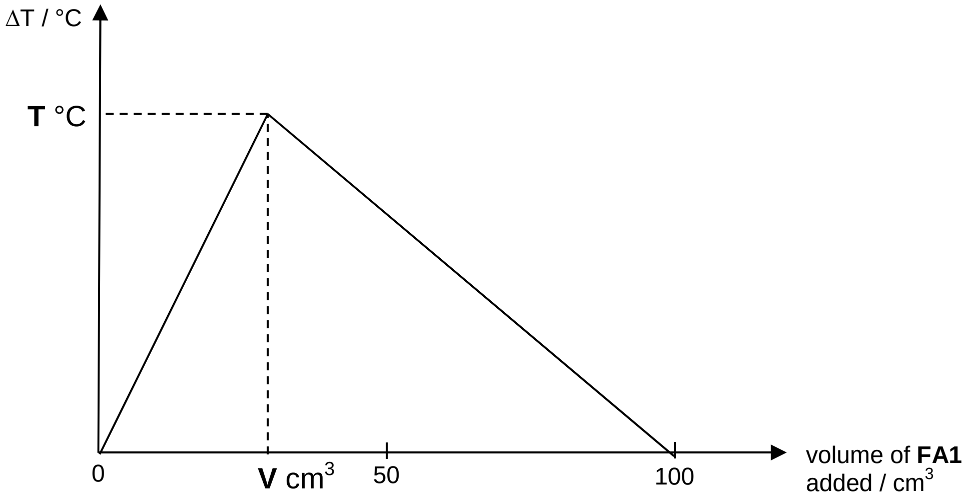
Paper 2: Structured Questions

1 **(a)** $\frac{1}{2}C_4H_2O_4 + NaOH \rightarrow \frac{1}{2}C_4O_4Na_2 + H_2O$

(b) (Due to the small temperature rise obtained), the use of the thermometer with divisions of 1 °C would lead to a large percentage error in the temperature rise (or formula for % error).
Or
the reading error of the thermometer with divisions of 1 °C is large **compared** to the small temperature rise obtained.
(do not accept "the reading error is large" without the comparison)

(c)
1. Using a 50 cm^3 / 100 cm^3 measuring cylinder, add 50 cm^3 of **FA1** solution into the styrofoam cup. Measure the initial temperature of this solution, T_1, using the thermometer.
2. Using another 50 cm^3 / 100 cm^3 measuring cylinder, measure out 50 cm^3 of **FA2**. Measure the initial temperature of this solution, T_2, using the thermometer.
3. The initial temperature, $T_{initial}$, is obtained by taking the weighted average of T_1 and T_2 (in this case it is just the mean of the two values as the volume of the solutions are the same).
4. Pour the **FA2** solution in the measuring the cylinder swiftly into the Styrofoam cup carefully without spillage, and cover the cup with the lid.
5. While stirring the resultant mixture gently with the thermometer, read the maximum temperature, T_{max}, reached (just before the temperature starts to drop).
6. The temperature rise for the experiment is just $T_{max} - T_{initial}$.

(d) & (e)



(f) **(i)** From the stoichiometry of the equation given in **(a)**,

$$\frac{n_{C_4H_2O_4}}{n_{NaOH}} = \frac{1}{2}, \Rightarrow n_{C_4H_2O_4} = \frac{1}{2} \times n_{NaOH}$$

$$\Rightarrow [C_4H_2O_4] \times V \times 10^{-3} = \frac{1}{2} \times 2.0 \times (100 - V) \times 10^{-3}$$

$$\Rightarrow [C_4H_2O_4] = \frac{100 - V}{V}$$

(ii) $q = 100 \times 4.2 \times T = 420T = 420 \times 10^{-3} T$ kJ

$$n_{reaction} = n_{H_2O}(formed) = \frac{1}{2} \times n_{NaOH} = \frac{1}{2} \times 2.0 \times (100 - V) \times 10^{-3} = (100 - V) \times 10^{-3} mol$$

$$\Rightarrow \Delta H_{neu} = -\frac{q}{n_{H_2O}} = -\frac{420 \times 10^{-3} T}{(100 - V) \times 10^{-3}} = -\frac{420T}{(100 - V)} kJmol^{-1}$$

(f) As the 2.0 mol dm^{-3} sodium hydroxide / squaric acid used in the experiment is corrosive, contact with skin should be avoided. Hence, gloves should be worn and droppers should be used when handling the sodium hydroxide solution.
Or
The Styrofoam cup which has a thermometer inside it, may topple and cause the thermometer to break, and the broken glass may cut somebody / the spilled mercury is highly toxic. Hence, the Styrofoam cup should be placed in a 250 cm^3 beaker, to prevent it from toppling over.

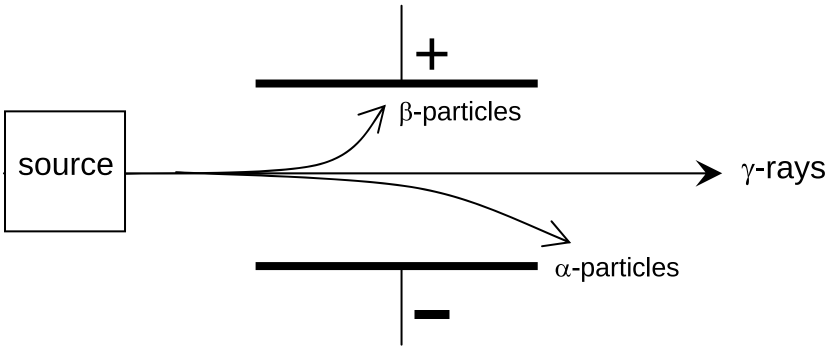
Or other relevant hazards identified and reasonable way of minimising risk.

2 **(a)** **(i)** Group V
The "big jump" of I.E. occurred between the fifth and sixth I.E., which indicates that there are five valence electrons.

(ii) Lower
(As the electronic configuration of element **L** is [Ne] 3s^2 3p^4), there are two electrons in one of the 3p orbital, and there will be inter-electronic repulsion, making the removal of one of the electrons easier than expected.

(iii) $JCl_4 + 2H_2O \rightarrow JO_2 + 4HCl$
or
$SiCl_4 + 2H_2O \rightarrow SiO_2 + 4HCl$
$KCl_3 + 3H_2O \rightarrow H_3KO_3 + 3HCl$
$KCl_5 + H_2O \rightarrow KOCl_3 + 2HCl$
$KCl_5 + 4H_2O \rightarrow H_3KO_4 + 5HCl$
or
$PCl_3 + 3H_2O \rightarrow H_3PO_3 + 3HCl$
$PCl_5 + H_2O \rightarrow POCl_3 + 2HCl$
$PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$
(either one of the equations)

(b) **(i)**



(α−particles towards negative plate, β-particles towards the negative plate, and α−particles deflected less than the β-particles)

(ii) protons: <u>**17**</u>, neutrons: <u>**19**</u>

(iii) *radiation **A***: <u>β-particles / electrons</u>
The conversion of ^{36}Cl to ^{36}Ar involves the increase of proton number by 1 units which corresponds to the loss of electron (β-particles) − a neutron converted into a proton and an electron

(iv) Assuming that the amount of chlorine-36 at the beginning is x, and there is no argon-36 present.

Time	^{36}Cl	^{36}Ar
0	x	0
$t_{1/2}$	$x/2$	$x/2$
$2t_{1/2}$	$x/4$	$3x/4$
$3t_{1/2}$	$x/8$	$7x/8$

time required = 3 x 300,000 = 900,000 years

(c) **(i)** titanium electrode: <u>$2Cl^- \rightarrow Cl_2 + 2e^-$</u>
steel electrode: <u>$2H^+ + 2e^- \rightarrow H_2$</u>

(ii)

$$n_{Cl_2} = \frac{1.00\times10^6}{71.0} = 14.1\times10^3\,mol$$

$$\Rightarrow n_{e^-} = 2\times n_{Cl_2} = 2\times14.1\times10^3 = 28.2\times10^3\,mol$$

$$Q = n_{e^-}\times F = 28.2\times10^3\times96500 = 2.72\times10^9\,C$$

$$\Rightarrow I = \frac{Q}{t} = \frac{2.72\times10^9}{24\times60\times60} = 31.5\times10^3\,A$$

(iii) equation: $Cl_2 + H_2 \rightarrow 2HCl$
potential safety hazard: In the presence of heat or uv light, the reaction may be explosive.

(iv) $Cl_2 + 2NaOH \rightarrow NaCl + NaClO + H_2O$

(d) When concentrated H_2SO_4 is added to solid halides, the first product is the hydrogen halide, which is seen as white fumes.
$KCl + H_2SO_4 \rightarrow HCl + KHSO_4$
$KBr + H_2SO_4 \rightarrow HBr + KHSO_4$ (either one)
(HCl has no further reaction with concentrated sulfuric acid)
HBr(g) will be further oxidised by the concentrated sulfuric acid to Br_2 which is seen as an orange fumes.
$2HBr + H_2SO_4 \rightarrow Br_2 + 2H_2O + SO_2$
This difference in the reactions is because HBr is a stonger reducing agent than HCl.

3 **(a)** **(i)** nucleophilic substitution / hydrolysis
electrophilic addition

(ii) cold concentrated H_2SO_4, followed by heating with H_2O
or
H_2O(g) (steam), H_3PO_4 catalyst, 300°C, 60 atm

(iii) **X** is formed from a five-membered ring (malic anhydride), and so the two R-groups must be on the same side of the carbon-carbon double bond (cis) – if not it will not be able to form the cyclic ring.

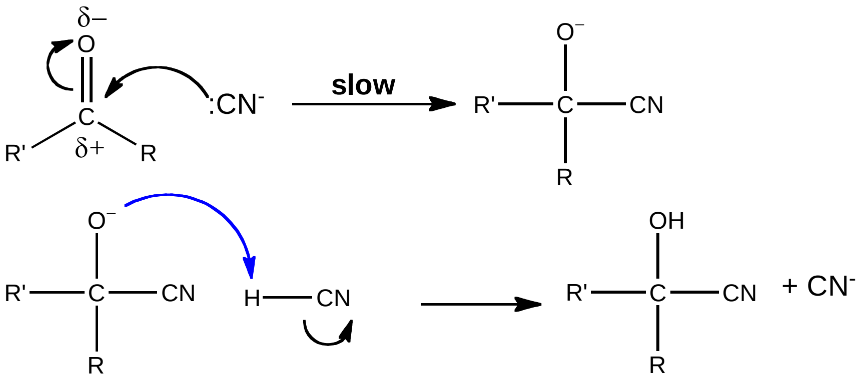
(iv) aqueous bromine / bromine in CCl_4
X will decolourise the brown aqueous bromine, no decolourisation for malic acid
or
$K_2Cr_2O_7$, dilute H_2SO_4, heat
malic acid will turn orange $K_2Cr_2O_7$ green, no colour change for **X**.
or
$KMnO_4$, dilute H_2SO_4, heat
X will decolour purple $KMnO_4$, and produce a gas that forms a white ppt with $Ca(OH)_2$; malic acid will only decolourise purple $KMnO_4$.

(b) **(i)** Comparing experiment 1 and 2,
When [HO_2CCH_2CHO] decreased by ¾ times, the rate also decreased by ¾ times
$\Rightarrow$ first order with respect to HO_2CCH_2CHO
Comparing experiment 1 and 3,
When [HCN] increased by 3 times, the rate remained the same
$\Rightarrow$ zero order with respect to HCN
Comparing experiment 2 and 4,
When [NaCN] increased by 8/5 times, the rate also increased by 8/5 times
$\Rightarrow$ first order with respect to NaCN
$\Rightarrow$rate equation: rate = k [HO_2CCH_2CHO] [NaCN]

(ii) As the volume of the solution is doubled, the concentration of HO_2CCH_2CHO and NaCN will both be halved.

⇒ the rate will be one-quarter of that for experiment 4 = 0.250

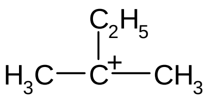
(iii) $NaCN(s) \rightarrow Na^{+}(aq) + CN^{-}(aq)$



(where R' = HO_2CCH_2-, and R is H−)
This mechanism is consistent with the rate equation from **(b)(i)**, because the rate determining step (slow step) involves one mole of HO_2CCH_2CHO and one mole of CN^{-}

(iv) The malic acid obtained from apple juice is optically pure / optically active / can rotate plane polarised light (not "chiral"), while the malic acid synthesised is a racemic mixture / optically inactive / does not rotate place polarised light. For step 2 of the mechanism, the planar arrangement of groups around the carbonyl carbon allows nucleophilic attack from either above or below the plane with equal probabilities, leading to equal concentration of both optical isomers.

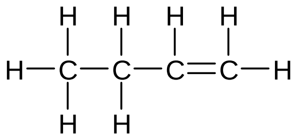
4 **(a)** **(i)**



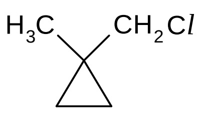
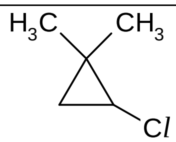
(ii) 2,2-dimethylbutanoic acid

(iii) number of electrons: <u>6</u>

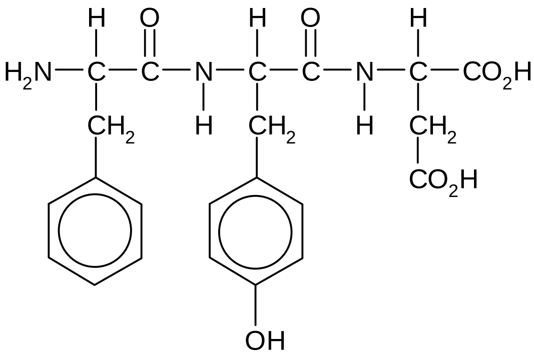
(iv)



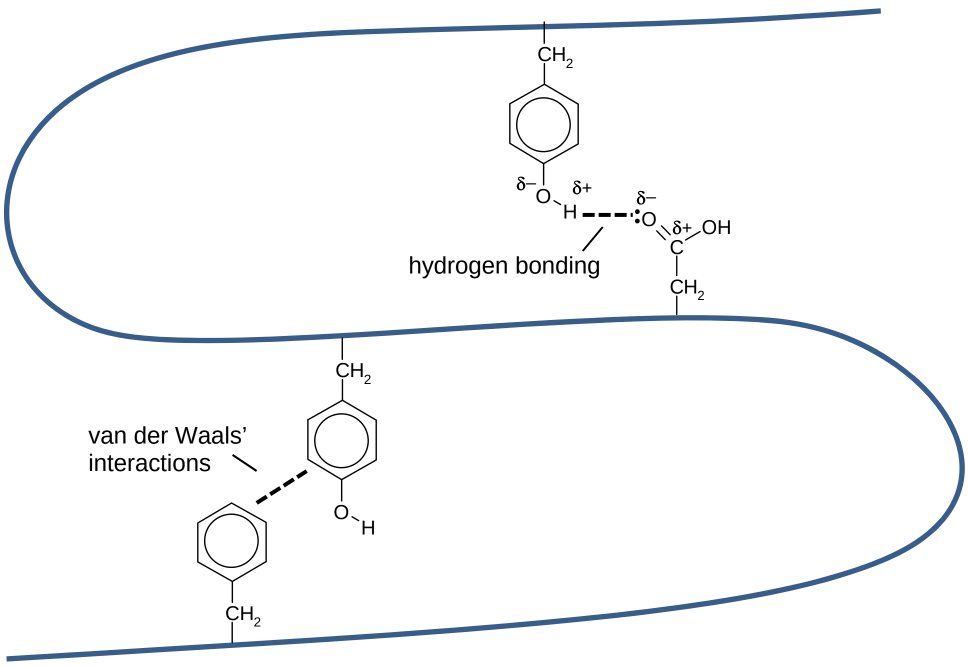
(b)

structural formula		
ratio	6	4

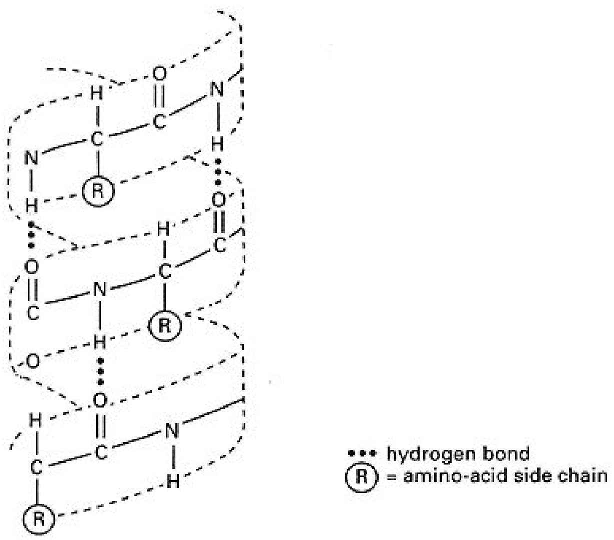
5 (a) (i)



(ii)



(b)



The α-helix is a regular coiled configuration of the polypeptide chain that resembles an extended spiral spring.
The structure is held in this fixed position (or stabilised) by hydrogen bonds between the N−H group of one amino acid residue and the fourth C=O group following it along the polypeptide chain (or shown in diagram)

(c) **(i)** CH_3OH, concentrated H_2SO_4, reflux

(ii) It is because aspartic acid has two carboxylic acid groups, and both may react with phenylalanine or methanol to form many unwanted products.

(d) **(i)** Ionic interactions exist between glycine molecules (as it forms zwitterions), while intermolecular hydrogen bonding exist between 2-hydroxyethanmide molecules.
A lot more energy is required to overcome the much stronger ionic interactions compared to the weaker intermolecular hydrogen bonding, and hence a higher temperature is required to melt glycine.

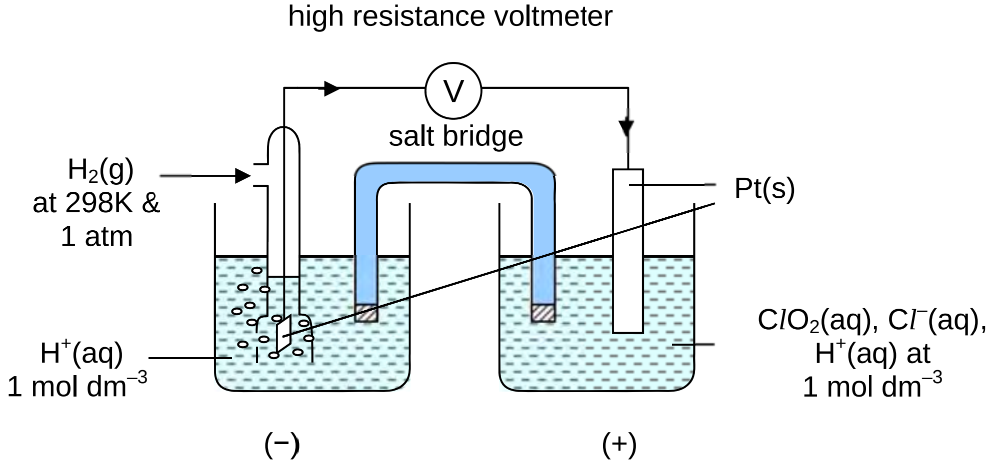
(ii) Cyclohexylamine is the strongest base as the cyclohexyl group is an electron donating group, and hence it intensifies the electron density of the lone pair of electrons on the nitrogen atom, making it more available for donation to a proton.
phenylamine is a weaker base than cyclohexylamine because the lone pair of electrons on the nitrogen atom interacts with the π-electron cloud of the benzene ring, and is therefore less available for donation.
N-phenylethanamide is the weakest base, as the lone pair of electrons on the nitrogen atom is delocalised over the carboxyl group in the amide, and is therefore the least available for donation.

(iii) Alkyl groups are 2,4-directing, and so the major products for nitration of methylbenzene is 1,2- and 1,4- substituted
For *tert*-butylbenzene, as the butyl group is very large and bulky, it will hinder the approach by the electrophile (NO_2^+) to attack at the carbon-2 position, and so the product is predominantly the 1,4-substituted product.

Paper 3: Free Response Questions

1 **(a)** **(i)** The standard electrode potential of an element is the potential difference between the element and its aqueous solution of 1.00 mol dm^{-3} relative to that of the standard hydrogen electrode at 1 atm and 298 K.

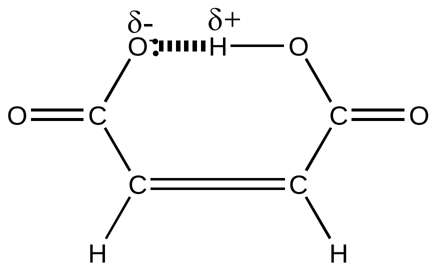
(ii)



(iii) $H_2S \rightarrow S + 2H^+ + 2e^-$
$2ClO_2(g) + 5H_2S(aq) \rightarrow 2Cl^-(aq) + 5S(s) + 4H_2O(l) + 2H^+(aq)$

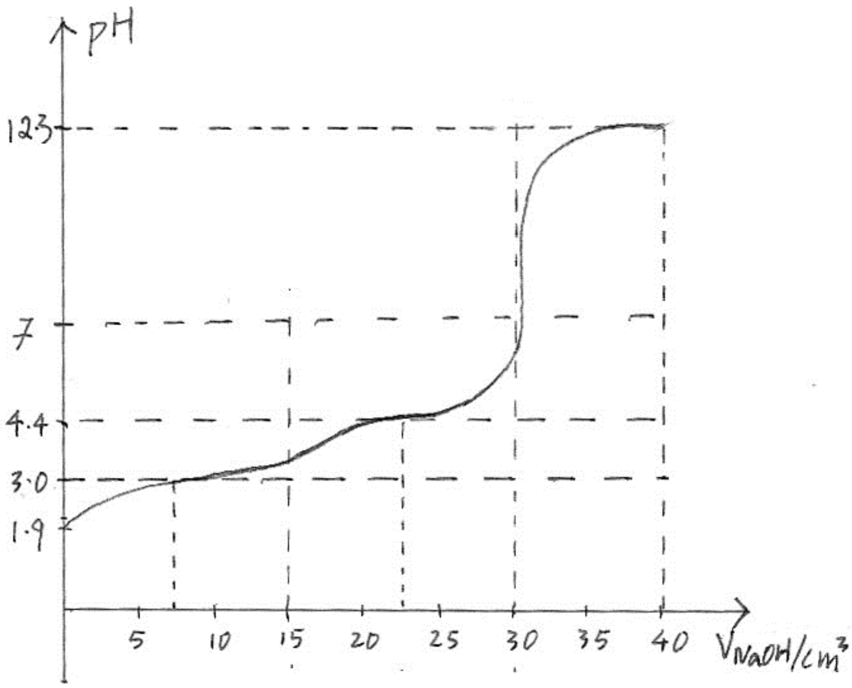
(b) **(i)** $^{-}OOC-CH=CH-COO^{-}$ is doubly negative charged and hence more likely to attract a H^+ to form $HOOC-CH=CH-COO^-$.
or
It is easier for a neutral molecule like $HOOC-CH=CH-COOH$ to donate a proton (H^+ ion) than from $HOOC-CH=CH-COO^-$ which is negatively charged.

(ii) The conjugate base of maleic acid is stabilised through intramolecular hydrogen bonds.



(iii) $[H^+] = \sqrt{0.15 \times (9.33 \times 10^{-4})} = 1.18 \times 10^{-2}\ \text{mol dm}^{-3}$
pH = 1.93

(iv)



(explanations for the **essential** points on the graph)

- initial pH on the titration curve = 1.93 {as calculated in part(iii)}
- volume of NaOH at the two end points = 15 cm^3 and 30 cm^3
  first end-point
  $HOOC-CH=CH-COOH + NaOH \rightarrow HOOC-CH=CH-COONa + H_2O$
  Volume of NaOH required = 0.0015 ÷ 0.10 = 0.015 dm^3 = 15 cm^3
  second end-point
  $HOOC-CH=CH-COONa + NaOH \rightarrow NaOOC-CH=CH-COONa + H_2O$
  Volume of NaOH required = 15.00 × 2 = 30 cm^3
- pH at the two half-neutralisation points = 3.03 and 4.44
  At 7.50 cm^3, [acid] = [mono-anion]. Hence pH = pK_1 = 3.03
  At 22.50 cm^3, [mono-anion] = [di-anion]. Hence pH = pK_2 = 4.44

(The following two points are not **essential** – i.e. bonus)

- pH at second equivalence point = 10.3

  $[OH^-] = \sqrt{(2.75 \times 10^{-10}) \times 0.0375} = 3.211 \times 10^{-6}$ mol dm^{-3}

  pOH = 5.49, pH = 8.51
- pH at 40 cm^3 NaOH added = 12.3

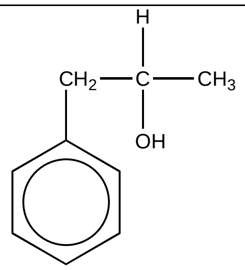
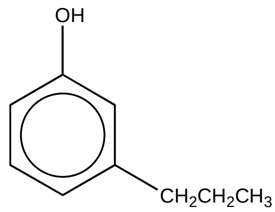
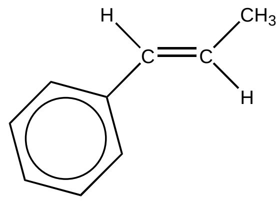
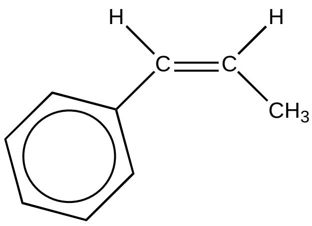
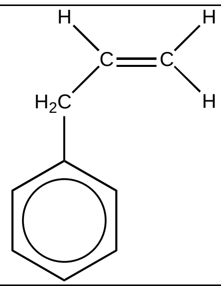
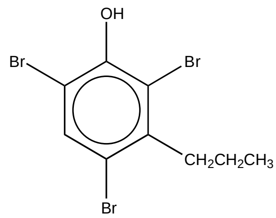
  No. of moles of excess OH^- = 1.00×10^{-3}

  $[OH^-] = (1.00 \times 10^{-3}) \div 0.05 = 2.00 \times 10^{-2}$ mol dm^{-3}

  pOH = 1.69, pH = 12.3

(d) **(i)** $C_9H_{12}O$

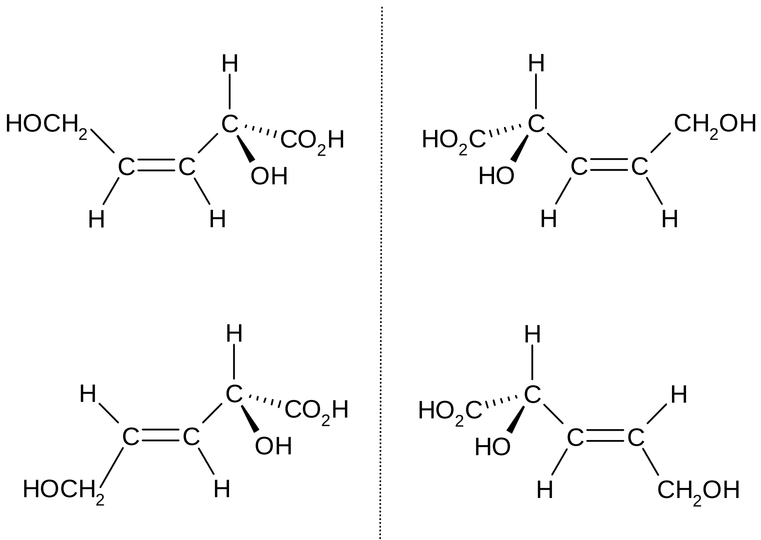
(ii)

		
A	**B**	**C**
		
D	**E**	**F**

2 **(a)** **(i)** dilute H_2SO_4, heat under reflux

(ii) Potassium manganate(VII) is a strong oxidising agent which will oxidise the oxalic acid to carbon dioxide and water instead.

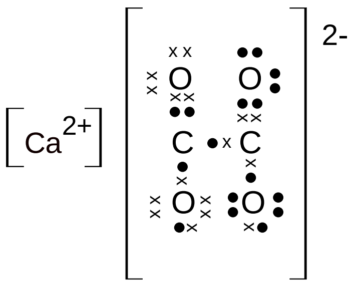
(iii) **Y** has 4 stereoisomers.



(b) **(i)** Carbon monoxide and carbon dioxide

(ii) They behave ideally under high temperature and low pressure.
At a high temperature, the gas particles have sufficient energy to overcome the intermolecular forces of attraction hence they are no longer attracted to one another.
At a low pressure, the gas particles are spaced far apart which makes their size negligible compared to the size of the container.

(c) **(i)**



(ii) The lone pair of electrons on the 2 O^- interacts with the π electrons of their adjacent C=O bond respectively, such that resonance occurs between the O–C–O atoms.

(d) **(i)** $K_{sp}(CaC_2O_4) = [Ca^{2+}][C_2O_4^{2-}]$ unit: $mol^2\ dm^{-6}$

(ii) Amount of $CaCl_2$ added = 0.00500 mol
Amount of $Na_2C_2O_4$ added = 0.0300 mol
As 1 mole of Ca^{2+} reacts with 1 moles of $C_2O_4^{2-}$ to form CaC_2O_4,
$C_2O_4^{2-}$ is in excess.
Amount of CaC_2O_4 formed = 5.00×10^{-3} mol

(iii) Amount of excess $C_2O_4^{2-}$ = 0.0300 – 0.00500 = 0.0250 mol
$[C_2O_4^{2-}]_{\text{left over}} = 0.0250 \div 0.150 = 0.167\ mol\ dm^{-3}$

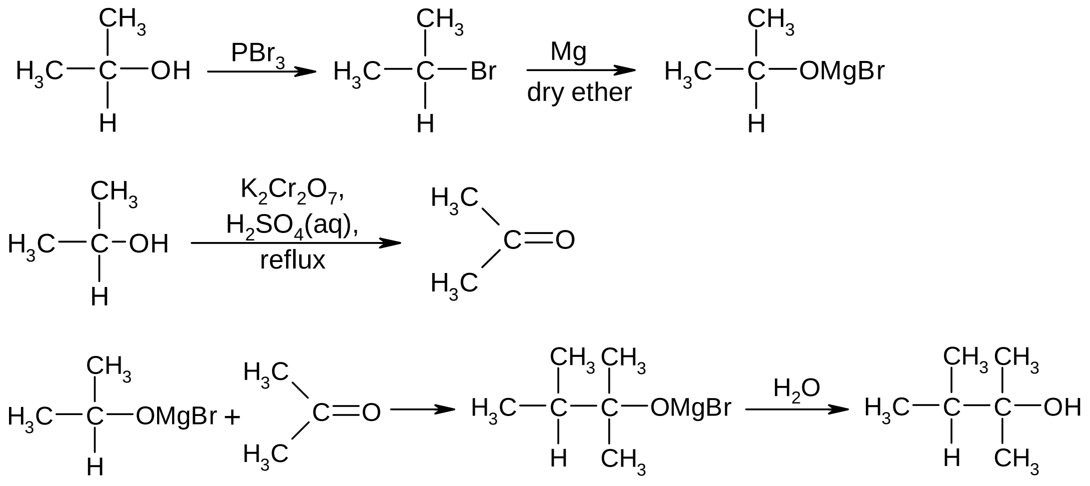
(iv) Let x be the solubility of CaC_2O_4.
$[Ca^{2+}] = x$ and $[C_2O_4^{2-}] = (x + 0.167)$
$2.32 \times 10^{-9} = (x)(x + 0.167)$, since x is very small, $(x + 0.167) \approx 0.167$,
$x = 1.39 \times 10^{-8}\ mol\ dm^{-3}$

(v) Not possible. As shown in part **(iv)**, there will still be $[Ca^{2+}]$ left over in the solution, after 100 cm^3 of $Na_2C_2O_4$ has been added.

(e) Mg^{2+} is smaller than Ca^{2+} and thus has a higher charge density and polarising power. Hence Mg^{2+} distorts the electron cloud of $C_2O_4^{2-}$ more significantly, and thus weakens the C–O bond to a greater extent compared to Ca^{2+}.
$\Rightarrow$ MgC_2O_4 decomposes at a lower temperature to form the metal oxide.

3 (a) (i) **I** : nucleophilic addition
II : hydrolysis

(ii)



(b) (i)

Number of moles of HC*l* = $\frac{15}{1000}$ x 0.200 = 3.00 x 10^{-3} mol

Number of moles of NH_3 = 3.00 x 10^{-3} mol

Number of moles of NH_2CONH_2 = ½ (3.00 x 10^{-3})
= 1.50 x 10^{-3} mol

Mass of nitrogen in urea = 2(1.50 x 10^{-3}) x 14 = 0.0420 g

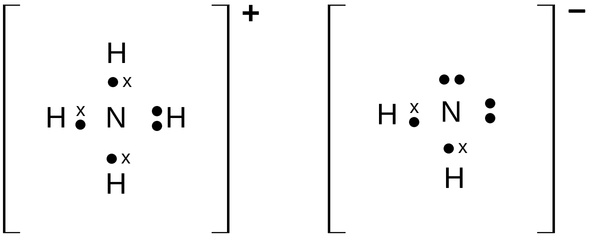
% of nitrogen in fertiliser = $\frac{0.0420}{0.100}$ x 100 = 42.0 %

(ii) $NH_2CONH_2 + H_2O \rightarrow 2NH_3 + CO_2$

(iii) Mg_3N_2 will react with the water in the soil to form $Mg(OH)_2$ and NH_3 (given in earlier part of question), the $Mg(OH)_2$ produced would increase the pH of the soil, making the soil unsuitable for plant growth.

(c) (i) $2NH_3 \rightleftharpoons NH_4^+ + NH_2^-$

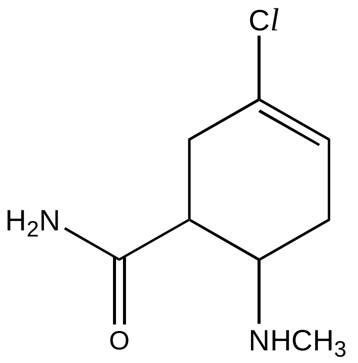
(ii)



109° 105°

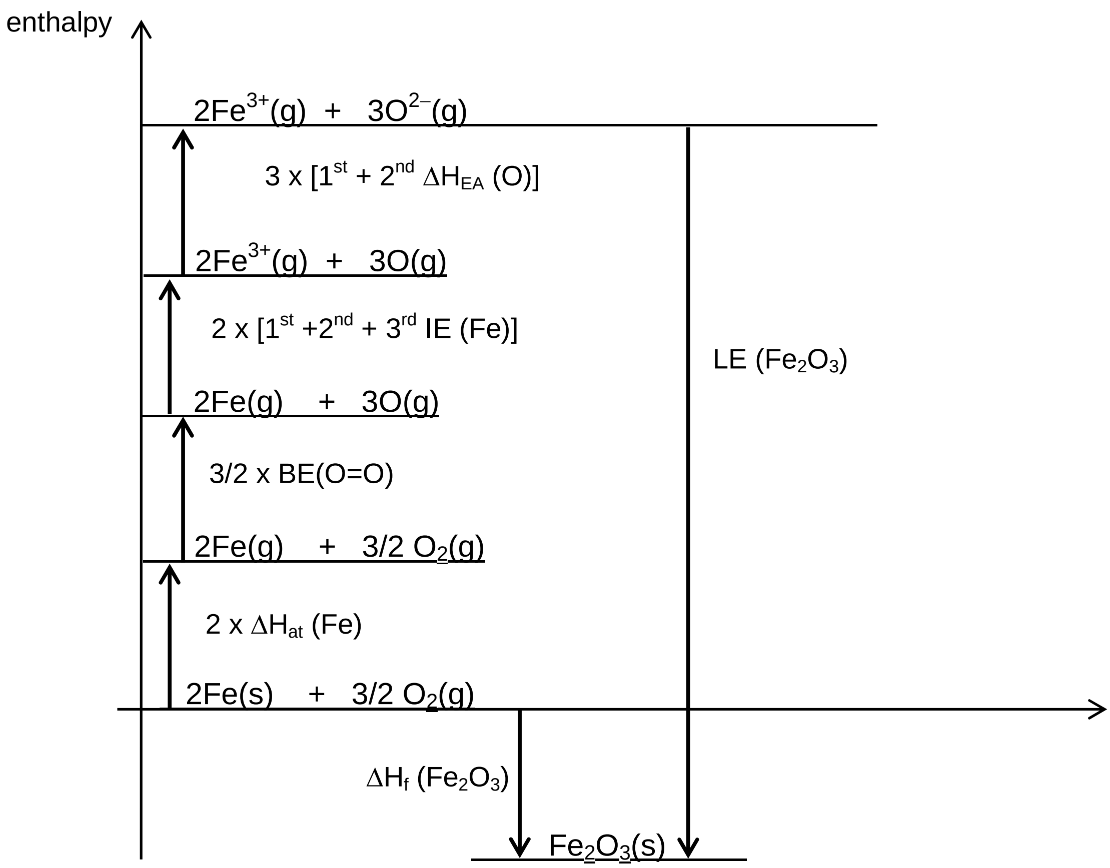
(d) **(i)** **III**: ethanolic CH_3Cl, heat under reflux

H:



(ii) There is overlapping of the lone pair of electrons on the chlorine atom with the π electron cloud of the alkene carbon atoms, resulting in partial double bond character between the chlorine and its immediate carbon atoms, which is strong and difficult to break.

4 **(a)** **(i)**



$\Delta H_f(Fe_2O_3) = 2\Delta H_{at}(Fe) + 2[1^{st}\ IE + 2^{nd}\ IE + 3^{rd}\ IE(Fe)] + \frac{3}{2}BE(O{=}O)$

$\qquad + 3[1^{st}\ \&\ 2^{nd}\ EA\ (O)] + LE(Fe_2O_3)$

$-826 \quad = 2(414) + 2(762+1560+2960) + \frac{3}{2}(496) + 3(703) + LE(Fe_2O_3)$

$LE\ (Fe_2O_3) = -1.51 \times 10^4\ kJ\ mol^{-1}$

(ii)

cation	ionic radius / nm
Fe^{3+}	0.064
Al^{3+}	0.050

Given that $|\text{lattice energy}| \propto \left|\dfrac{q_+ \times q_-}{r_+ + r_-}\right|$, since Al^{3+} is smaller than Fe^{3+}, the magnitude of lattice energy for Al_2O_3 is greater (since the denominator of the fraction is smaller), i.e. there is a stronger attraction between Al^{3+} and O^{2-} ions which requires more energy to break.

(iii) $Al_2O_3 + 6H^+ \rightarrow 2Al^{3+} + 3H_2O$ or $Fe_2O_3 + 6H^+ \rightarrow 2Fe^{3+} + 3H_2O$

$Al_2O_3 + 2OH^- + 3H_2O \rightarrow 2Al(OH)_4^-$

(b) (i) Iron contains partially filled 3d orbitals (and other energetically accessible orbitals) which can form weak bonds / interactions with the reactant molecules.

(ii)
1. Reactant molecules H_2 and CO are adsorbed onto the surface of iron catalyst.
2. Weakening of the H–H and C≡O bonds in the reactant molecules occur and this lowers the activation energy of the reaction, which is followed by the formation of C–H and O–H bonds in product molecules.
3. Product molecules are desorbed from the surface of iron catalyst.

(iii) The physical state of the product hydrocarbon changes from gas to liquid / solid with an increasing carbon chain length.
As entropy of a liquid / solid is lower than that of a gas, hence the decrease in the entropy of the reaction becomes more significant as the physical state of the product hydrocarbon changes from gas to liquid / solid.

(c) (i) The quaternary structure comprises 4 polypeptides (2α and 2β subunits), each with its own haem group, and every haem group consists of a central iron(II).
The structure is maintained by intermolecular van der Waals' forces, hydrogen bonding and ionic interactions between the R-groups of the amino acid residues. (note: there are no disulfide bridges present within each subunit)

(ii) In the lungs where concentration of oxygen is higher, equilibrium for the above reaction will shift to the right to form oxyhaemoglobin molecules, $Hb(O_2)_4$, which are carried to the vital organs in our human bodies.
In the vital organs where concentration of oxygen is lower, equilibrium will then shift to the left to release the oxygen molecules and form deoxyhaemoglobin molecules, Hb, which are then transported back to the lungs.

(iii) As carbon monoxide is a stronger ligand than oxygen and water, it binds irreversibly to Fe^{2+} in haemoglobin molecules which causes them to lose their ability to transport oxygen in our bodies.

(iv) Change in the structure of the protein such that it loses its biological activity.
In basic medium, the ionic interactions between oppositely charged R groups (–COO^- and –NH_3^+) are disrupted, which causes the protein chain to unfold.

5 (a) (i) A transition element is a d-block element that forms at least one or more stable ions with a partially filled d-subshell.

(ii) In a complex, the presence of ligands causes a splitting of the d-orbitals into two sets with different energy levels. The difference in energy between the d and d* orbitals, ΔE, falls within the energy (wavelength) range of visible light.
When exposed to light, an electron from the lower energy d orbitals can absorb certain wavelengths of visible light and promotes to the higher energy d* orbitals.
The colour of the complex observed will therefore be the mixture of wavelengths of visible light that are not absorbed i.e. the complimentary colour is emitted.

(b) (i) The blue solution turns yellow.

$$\underset{\text{blue}}{[Cu(H_2O)_6]^{2+}} + 4Cl^- \rightleftharpoons \underset{\text{yellow}}{[CuCl_4]^{2-}} + 6H_2O$$

(ii) The yellow solution will turn deep blue due to ligand exchange reaction.

$$\underset{\text{yellow}}{[CuCl_4]^{2-}} + 4NH_3 \rightleftharpoons \underset{\text{deep blue}}{[Cu(NH_3)_4]^{2+}} + 4Cl^-$$

(c) (i) Cu^+ disproportionates to Cu (pink solid) and Cu^{2+} (blue solution).
$2Cu^+(aq) \rightarrow Cu(s) + Cu^{2+}(aq)$
$E^{\ominus}_{cell} = +0.37V$

(ii) $2Cu^{2+}(aq) + 4I^-(aq) \rightarrow 2CuI(s) + I_2(aq)$
$E^{\ominus}_{cell} = -0.39V$
Since the $E^{\ominus}_{cell}$ is negative, the reaction is not feasible.

(iii) The precipitation of CuI reduces the concentration $Cu^+(aq)$, promoting the reduction of $Cu^{2+}(aq)$ to $Cu^+(aq)$ and hence making the reduction potential of Cu^{2+}/Cu^+ more positive such that $E^{\ominus}_{cell}$ is positive and the redox reaction becomes spontaneous/feasible.
Or
The reaction is not carried out under standard conditions.

(d) (i) No. of moles of $O_2 = 20.4 \div 24000 = 8.50 \times 10^{-4}$

No. of moles of $H^+ = (0.1 \times 0.017) \times 2 = 3.40 \times 10^{-3}$

No. of moles of Cu $= (0.108 \div 63.5) \times 2 = 3.40 \times 10^{-3}$

(ii) No. of moles of **Z** $= 3.40 \times 10^{-3}$

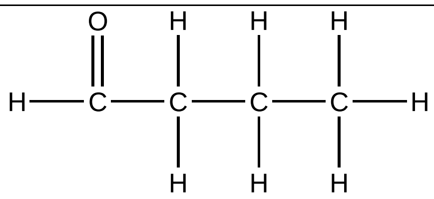
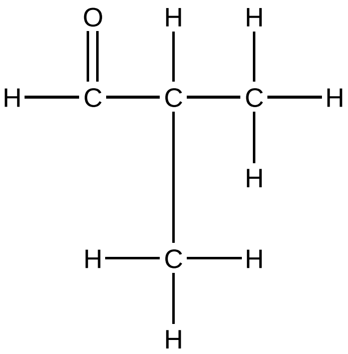
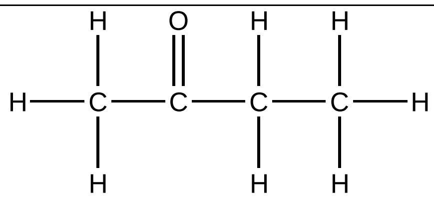
Relative formula mass of **Z** $= 1 \div 3.40 \times 10^{-3} = 294$

Formula of **Z**: K_3CuF_6

(iii) $3F_2 + 3KCl + CuCl_2 \rightarrow K_3CuF_6 + 2\tfrac{1}{2}Cl_2$

$4K_3CuF_6 + 2H_2O \rightarrow O_2 + 4HF + 4CuF_2 + 12KF$

(e)

 or 	
positive test	negative test

~ END OF SUGGESTED ANSWERS ~